RA8P1 E-READER

Processor schematic development
Power networks are being implemented from RA8P1-specific references.
This revision is not a completed schematic or a fabrication release.

Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

Remaining subsystem circuits follow epic #821 and issues #824–#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

Footprint and PCB qualification are deferred by scope.

RA8P1 IO allocation

File: mcu_interfaces.kicad_sch

RA8P1 internal core regulator

File: mcu_core_power.kicad_sch

RA8P1 IO and analog supplies

File: mcu_supply_power.kicad_sch

RA8P1 clocks, reset and debug

File: mcu_clocks_debug.kicad_sch

Sheet: /		
File: ereader_rev1.kicad_sch		
Title: RA8P1 e-reader – schematic index		
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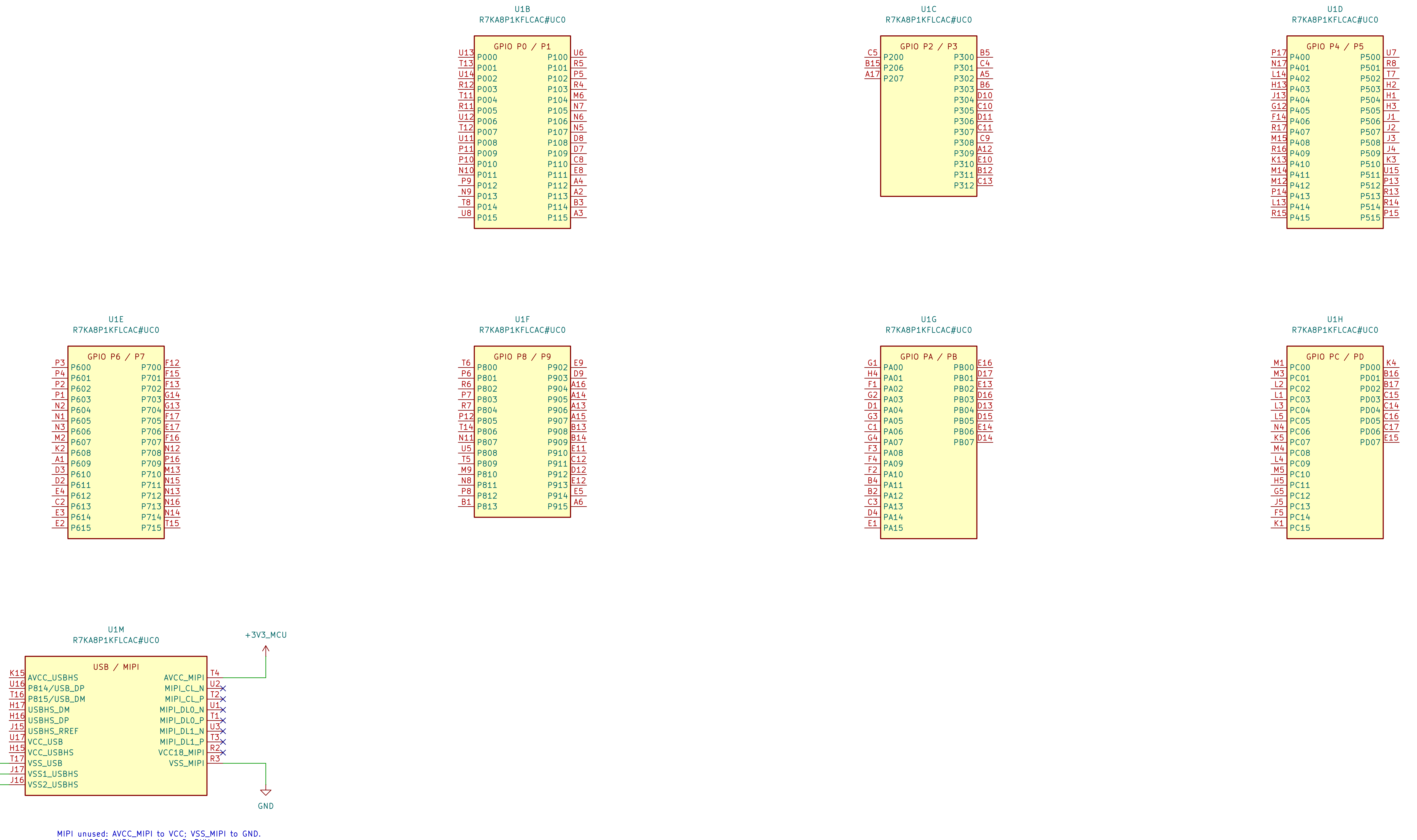
KiCad E.D.A. 10.0.5		Id: 1/5

DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D−PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.

RA8P1 e-reader

Sheet: /RA8P1 IO allocation/
File: mcu_interfaces.kicad_sch

Title: RA8P1 IO allocation

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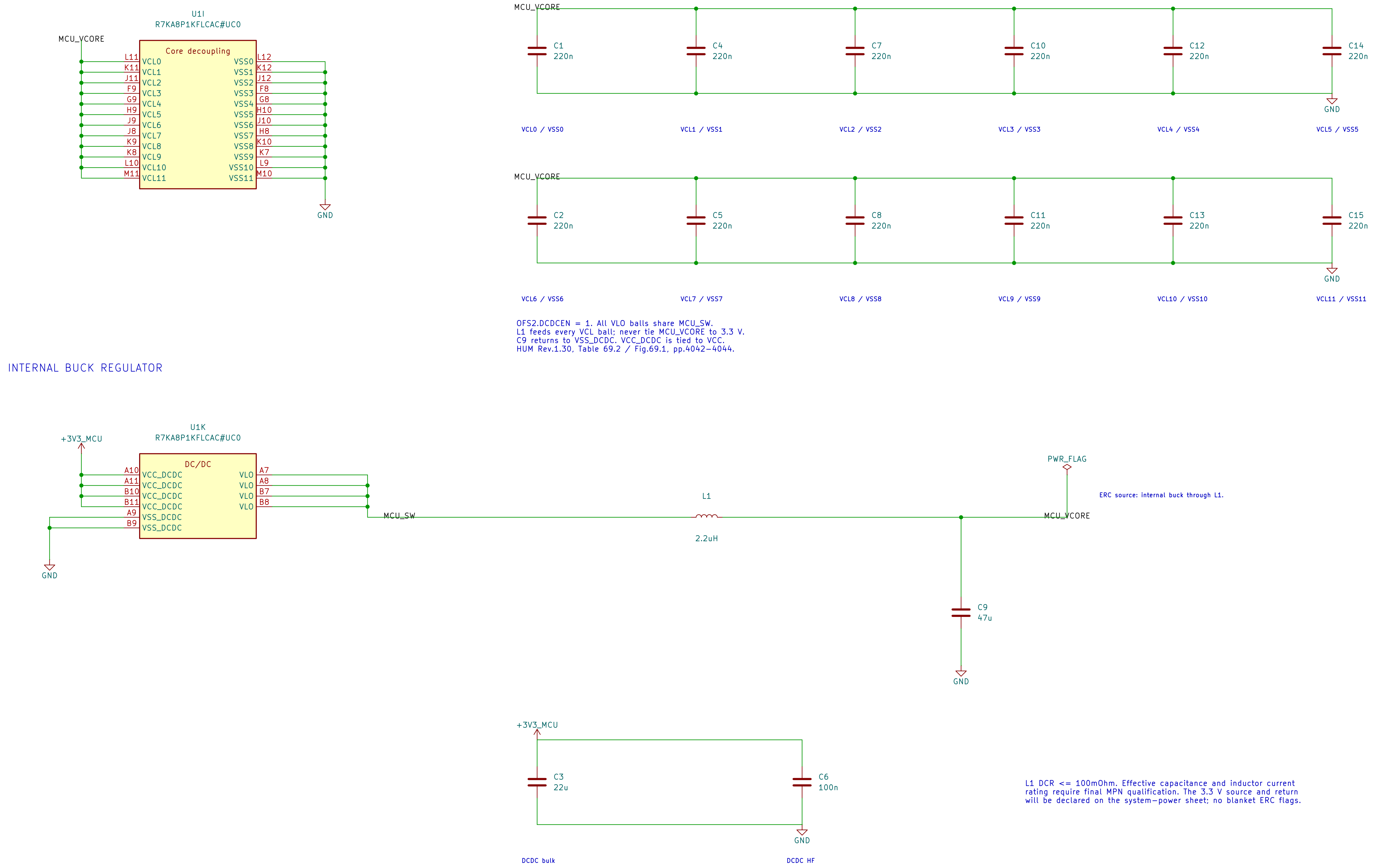
CORE DECOUPLING – 220 nF AT EACH VCL / VSS PAIR

INTERNAL BUCK REGULATOR



OFS2.DCDCEN = 1. All VLO balls share MCU_SW.
L1 feeds every VCL ball; never tie MCU_VCORE to 3.3 V.
C9 returns to VSS_DCDC. VCC_DCDC is tied to VCC.
HUM Rev.1.30, Table 69.2 / Fig.69.1, pp.4042–4044.

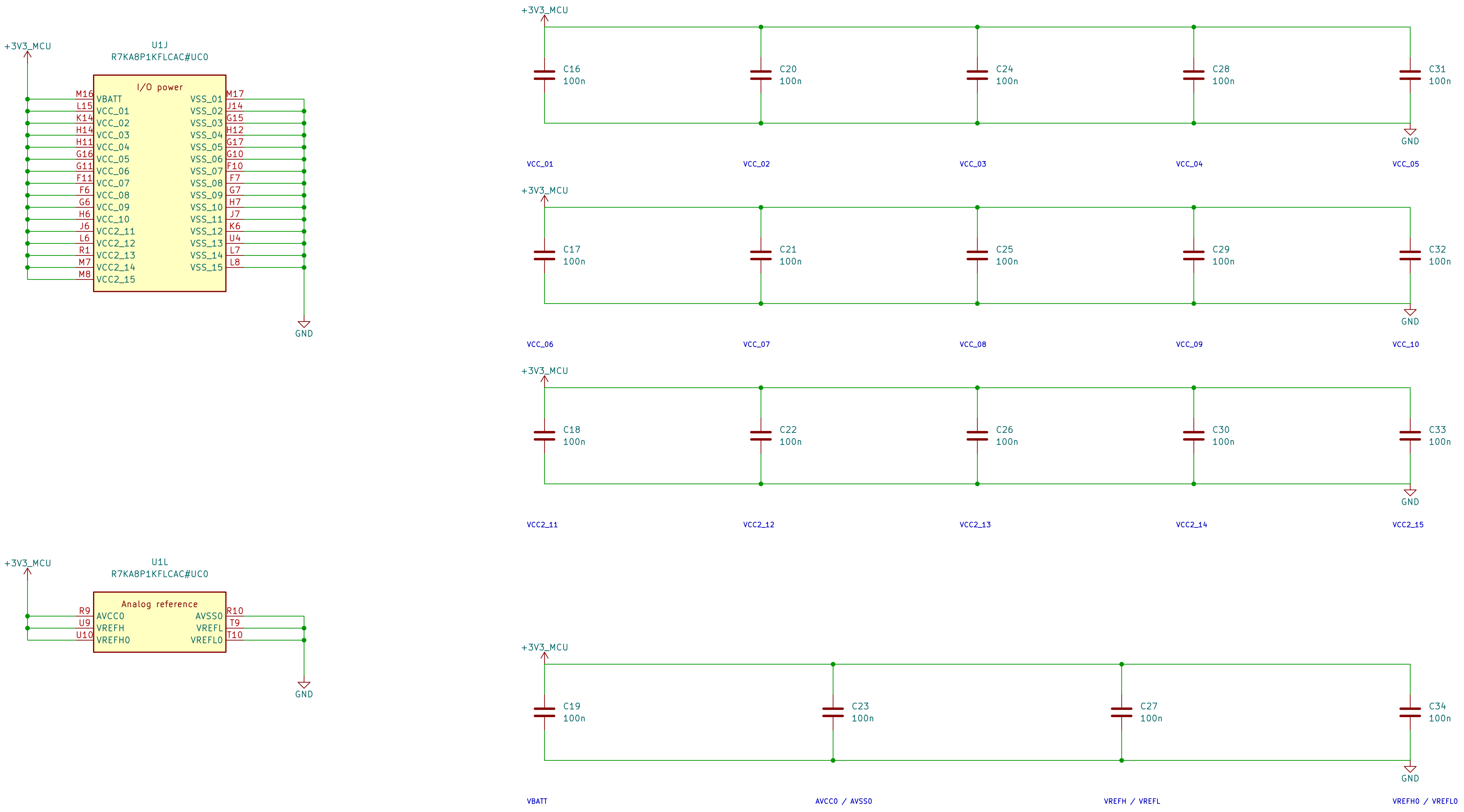
L1 DCR <= 100mOhm. Effective capacitance and inductor current rating require final MPN qualification. The 3.3 V source and return will be declared on the system-power sheet; no blanket ERC flags.

RA8P1 e-reader
Sheet: /RA8P1 internal core regulator/
File: mcu_core_power.kicad_sch

Title: RA8P1 internal core regulator

Size: A3	Date: 2026-09-05	**Rev: 0.3 DRAFT**
KiCad E.D.A. 10.0.5		Id: 3/5

I/O DECOUPLING — 100 nF AT EACH NUMBERED VCC / VSS PAIR



VCC and VCC2 use 3.3 V. All returns use one board ground plane.
VBATT follows the main rail; no separate backup-cell supply.
Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs.
USB/MIPI supplies remain on the interface sheet and are not yet wired.
HUM Rev.1.30: Table 69.2 p.4042; unused references: section 21.4 pp.861–862.
This is an electrical draft; capacitor MPN qualification remains open.

RA8P1 e-reader
Sheet: /RA8P1 IO and analog supplies/
File: mcu_supply_power.kicad_sch

Title: RA8P1 I/O and analog supplies

Size: A3	Date: 2026-09-05	**Rev: 0.3 DRAFT**
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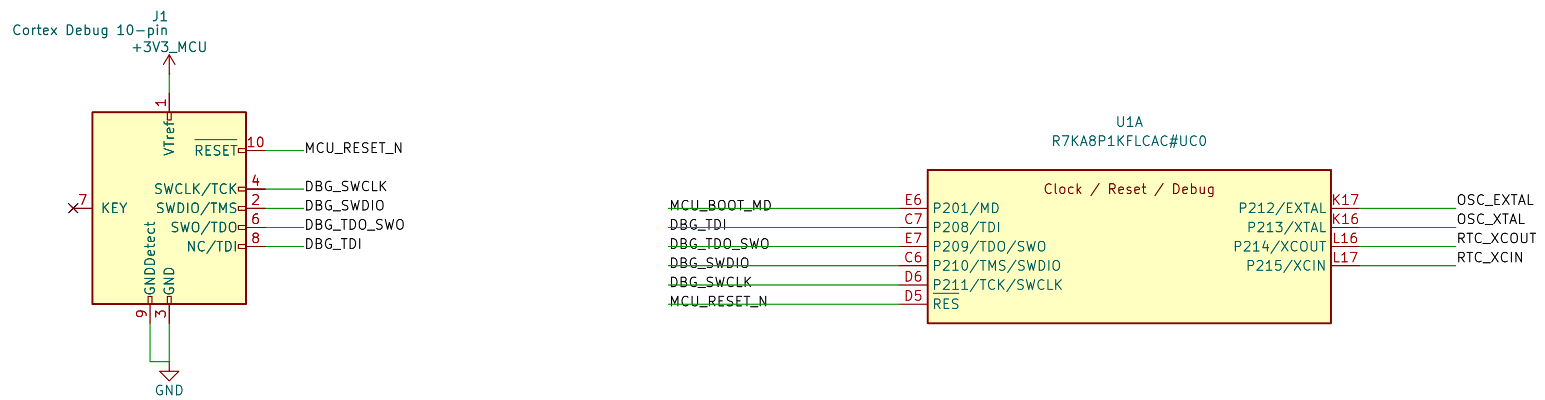

CLK-001: Y1 24 MHz / Murata XRCGB24M000F3M19R0
Crystal CL = 6 pF; effective Cstray = 2 pF ASSUMED.
CL = C35*C36/(C35+C36) + Cstray
C35 = C36 = 2*(6-2) = 8 pF (NP0, +/-0.5 pF).
Check: 8*8/(8+8) + 2 = 6 pF. At fixed stray: 5.75..6.25 pF.
Initial matching only: verify frequency, startup and drive <= 300 uW.
Full derivation + Python check: ../design/clock_calculations.md

CLK-002: Y2 ABS07-LR-32.768KHZ-6-1-T
Crystal CL=6 pF; Cstray=4 pF ASSUMED.
CL=C37*C38/(C37+C38)+Cstray
C37=C38=2*(6-4)=4 pF, +/-0.25 pF NP0.
Check: 4*4/(4+4)+4=6 pF.
Fixed-stray tolerance: 5.875..6.125 pF.
Initial mode: SODRV=10b; SOSEL=0.
Qualify startup, frequency, drive <=0.5 uW.
Full math/Python: ../design/clock_calculations.md

NO capacitor on RES: it must track VCC for internal POR.
External reset drivers must be open-drain/open-collector.
Hold RESET low >= 3 ms, then release after VCC is valid.
Hold BOOT during external reset release for SCI/USBFS mode.
Normal MD high; SWD/JTAG boot requests occur during RES low.
Device lifecycle / authentication restrictions still apply.

References: RA8P1 HUM Rev.1.30 sections 2.5, 4.3-4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RA8P1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.

Sheet: /RA8P1 clocks, reset and debug/
File: mcu_clocks_debug.kicad_sch

Title: RA8P1 clocks, reset and debug

Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
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